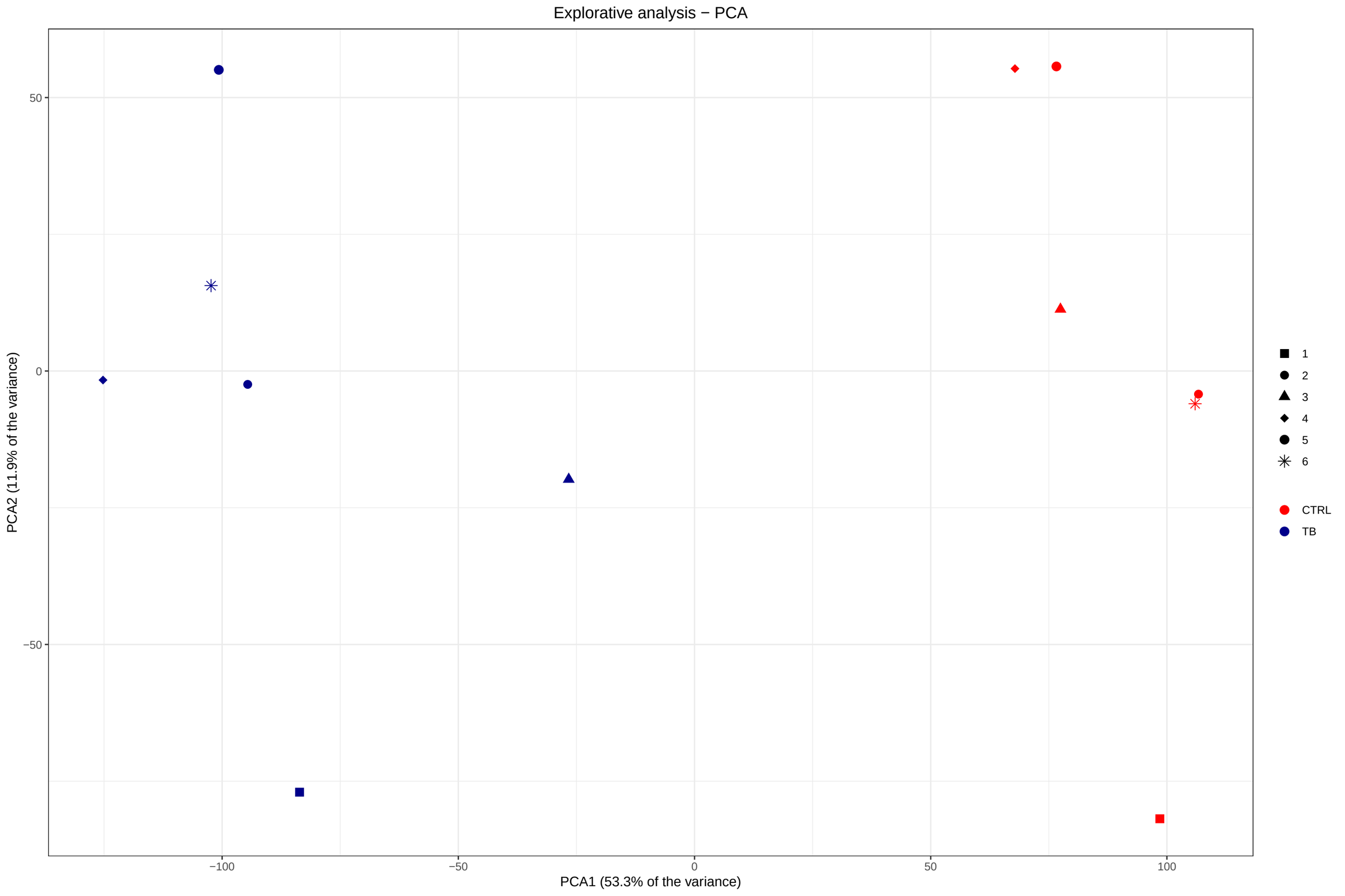
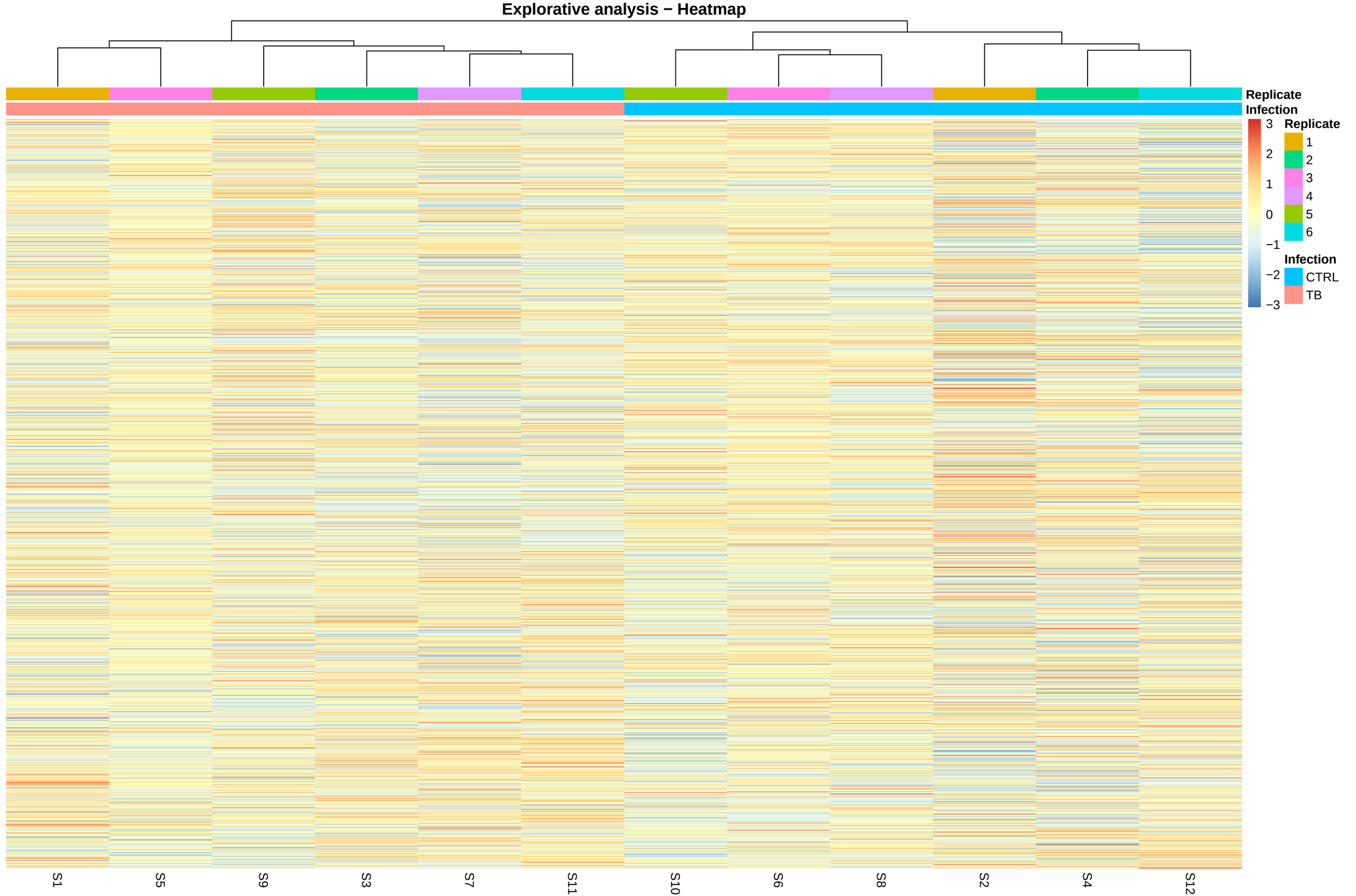
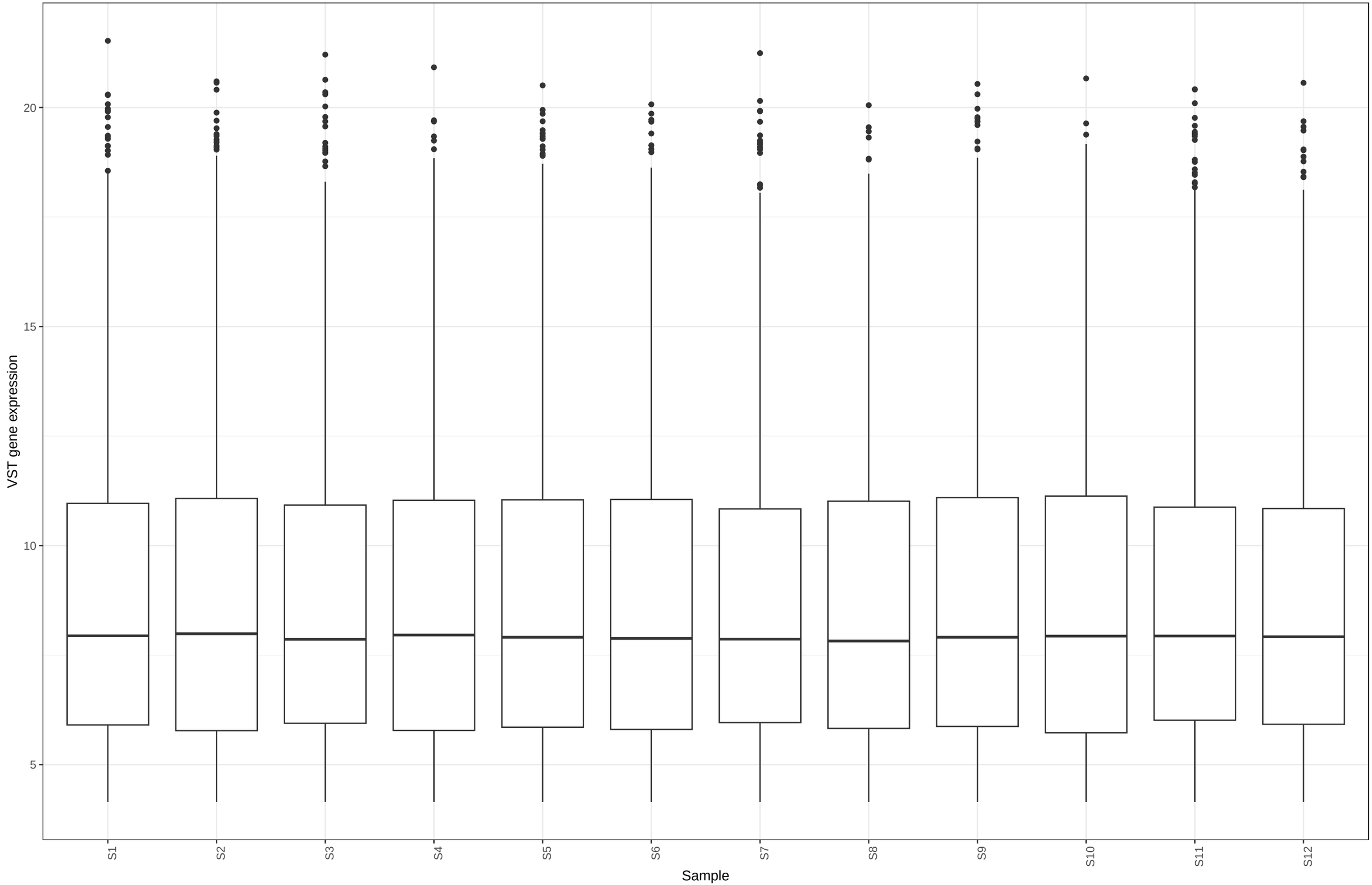
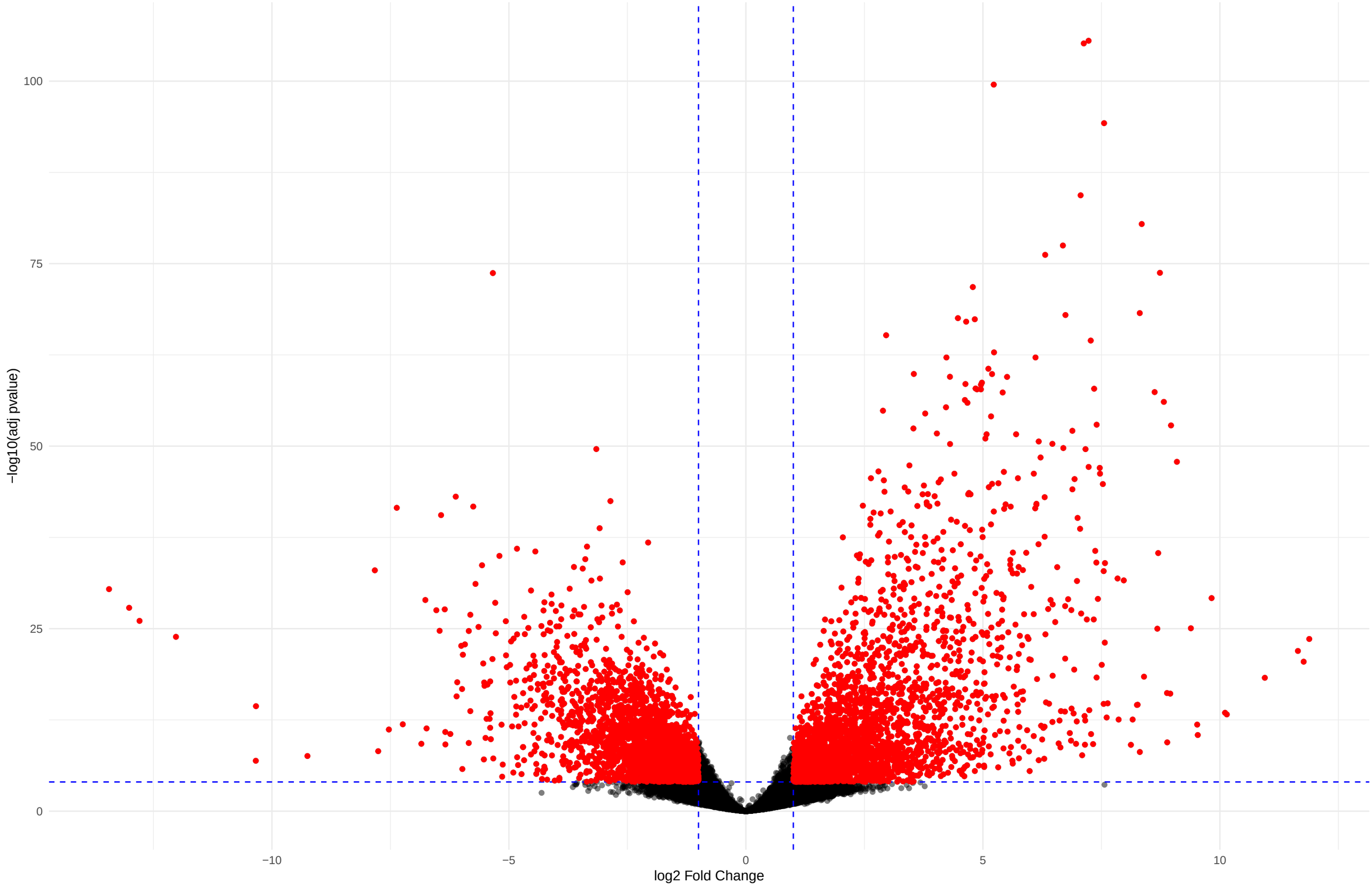

Explorative analysis − PCA
PCA2 (11.9% of the variance)
PCA1 (53.3% of the variance)
50
0
−50
−100
−50
0
50
100
1
2
3
4
5
6
CTRL
TB

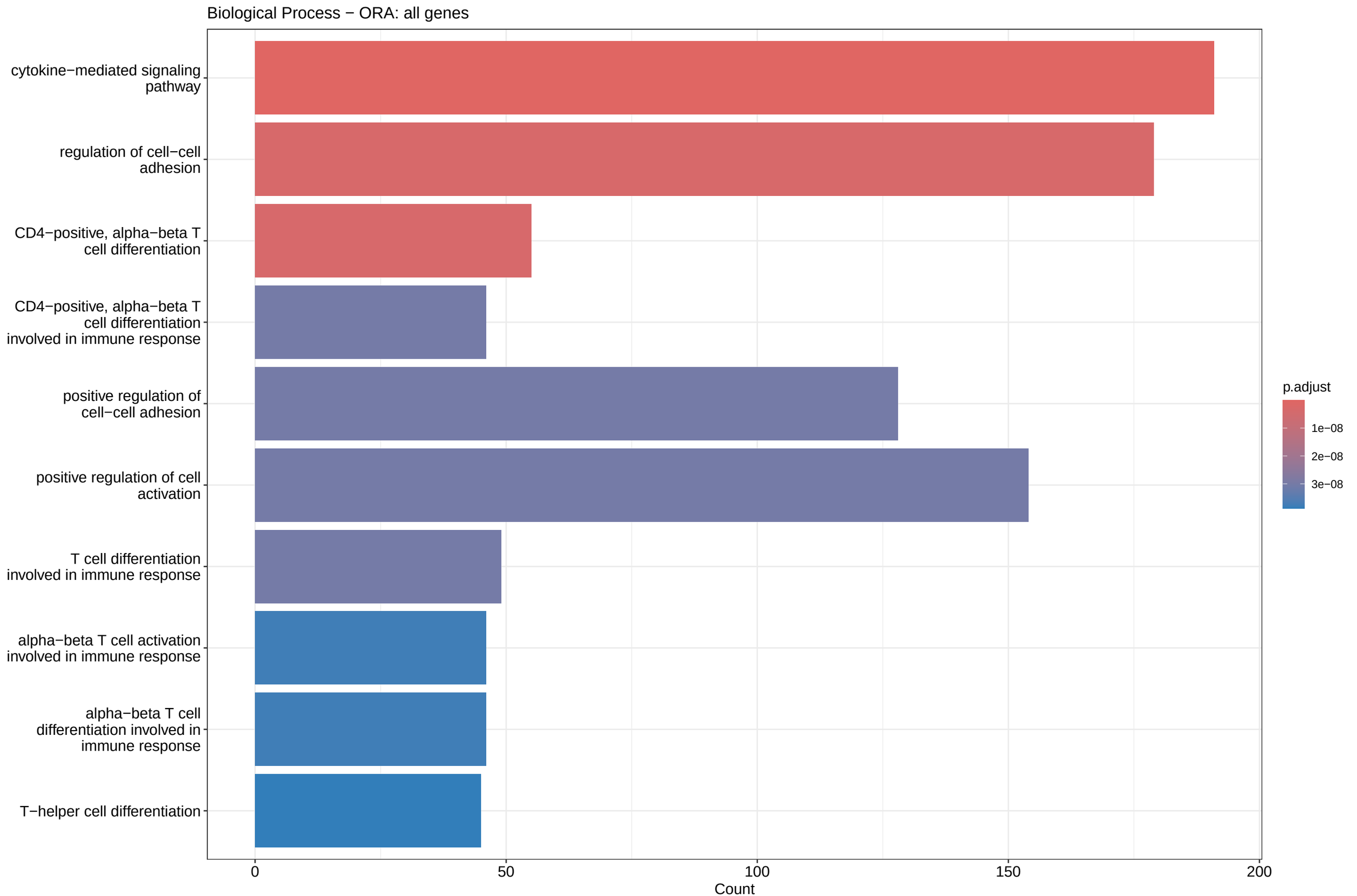

Explorative analysis − Heatmap
Replicate
Infection
3
2
1
0
−1
−2
−3
Replicate
1
2
3
4
5
6
Infection
CTRL
TB
S1
S5
S9
S3
S7
S11
S10
S6
S8
S2
S4
S12

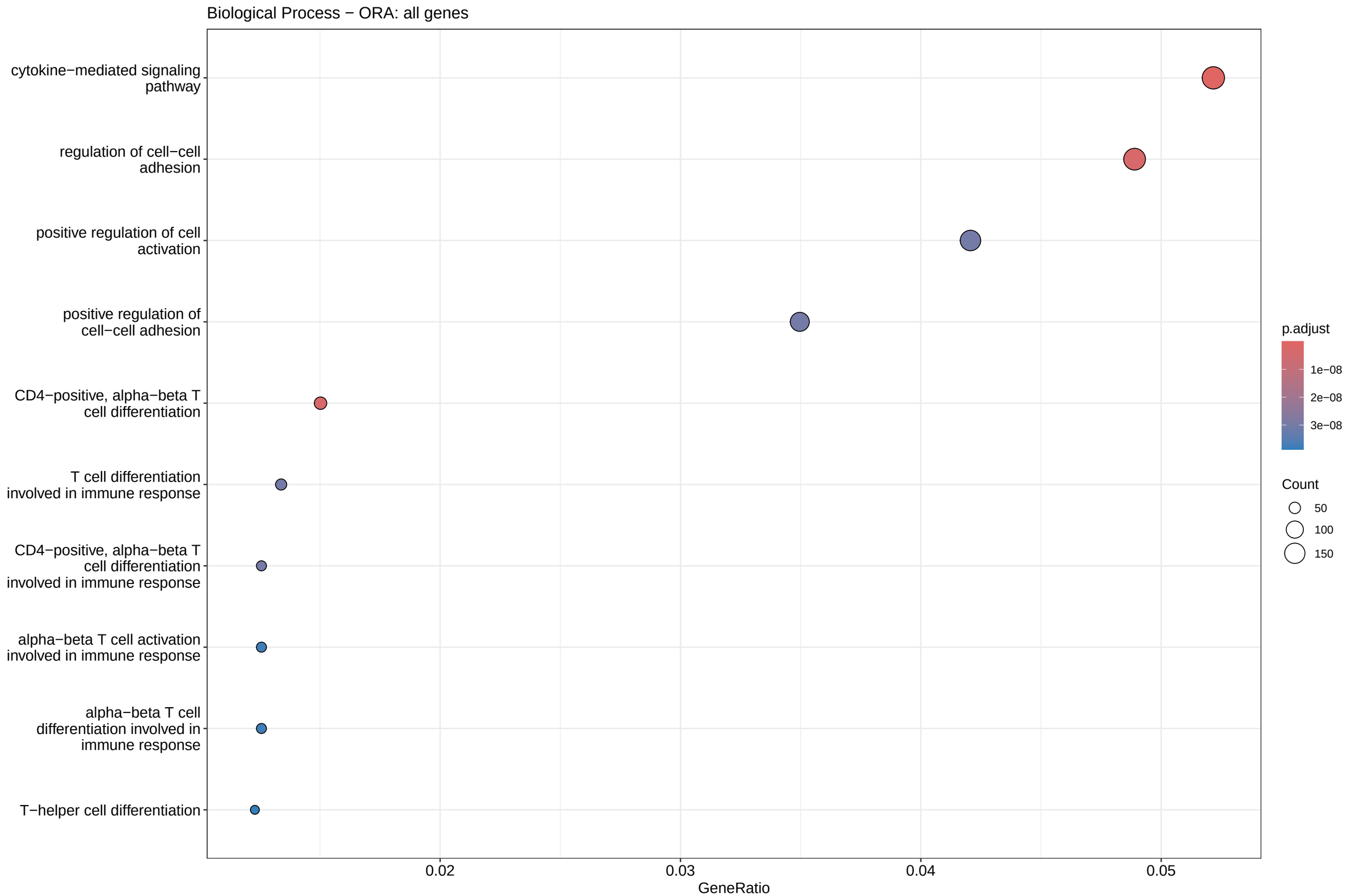
Explorative analysis − Expression boxplots

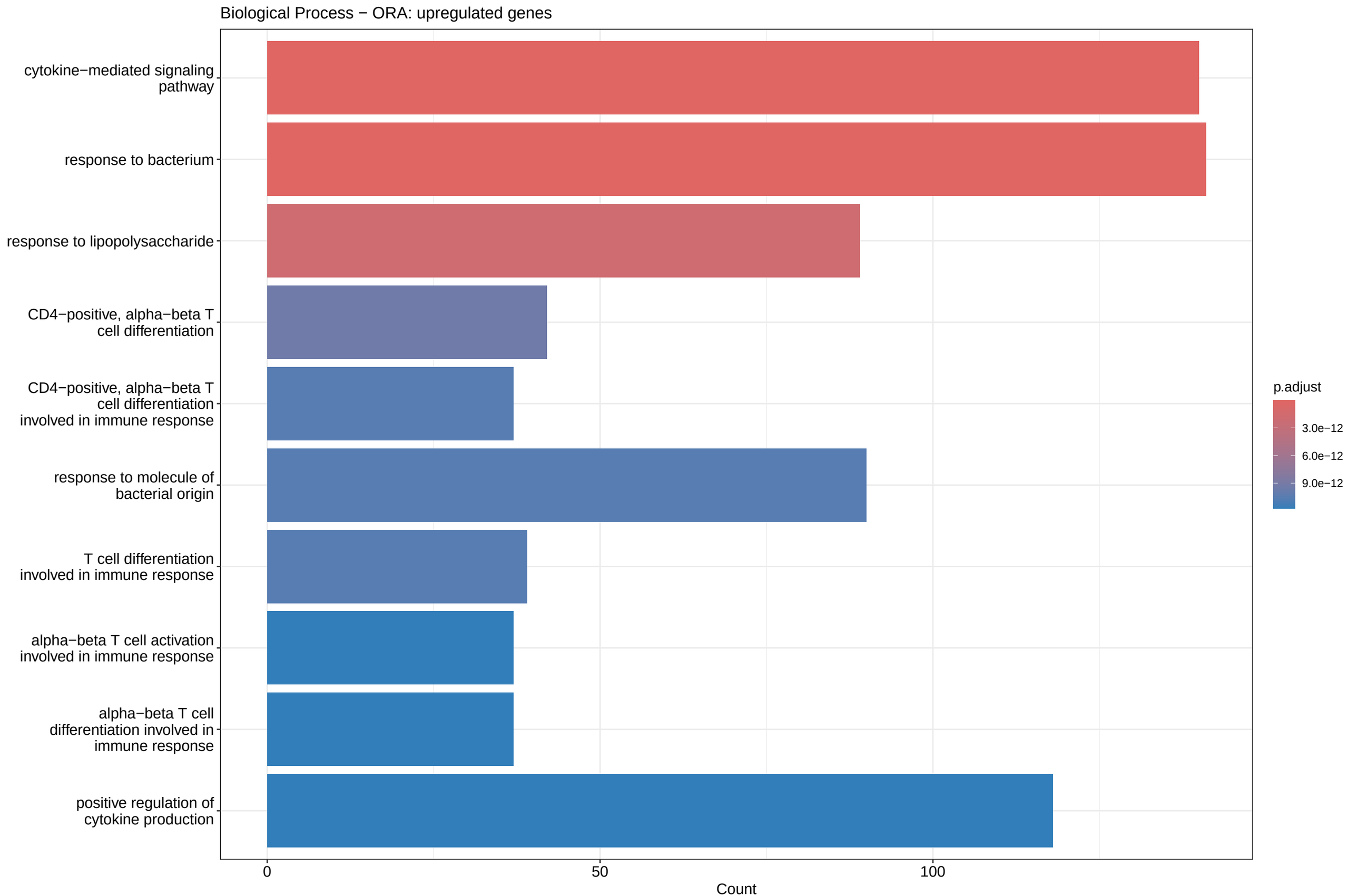


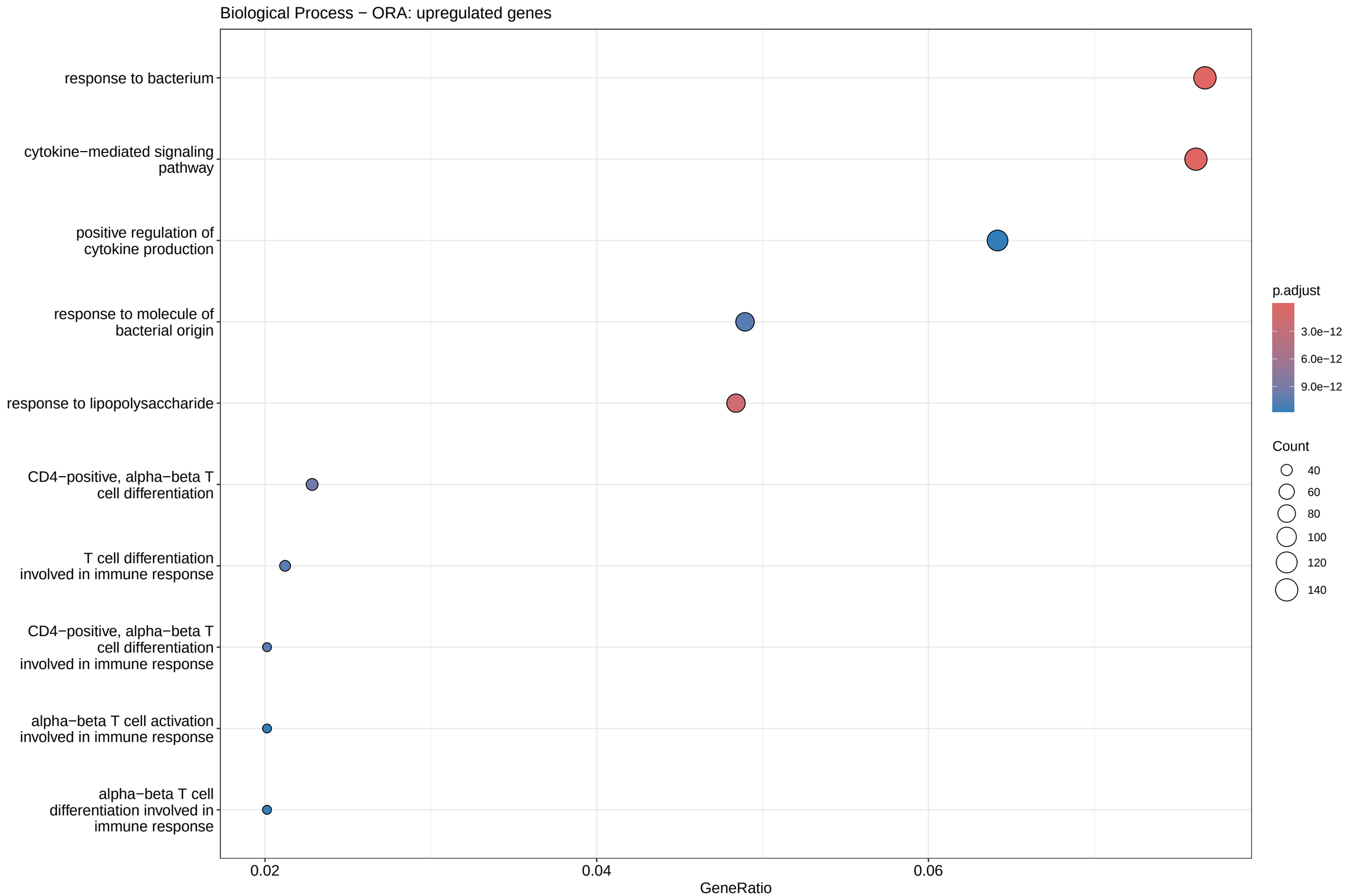
Volcano plot: infection

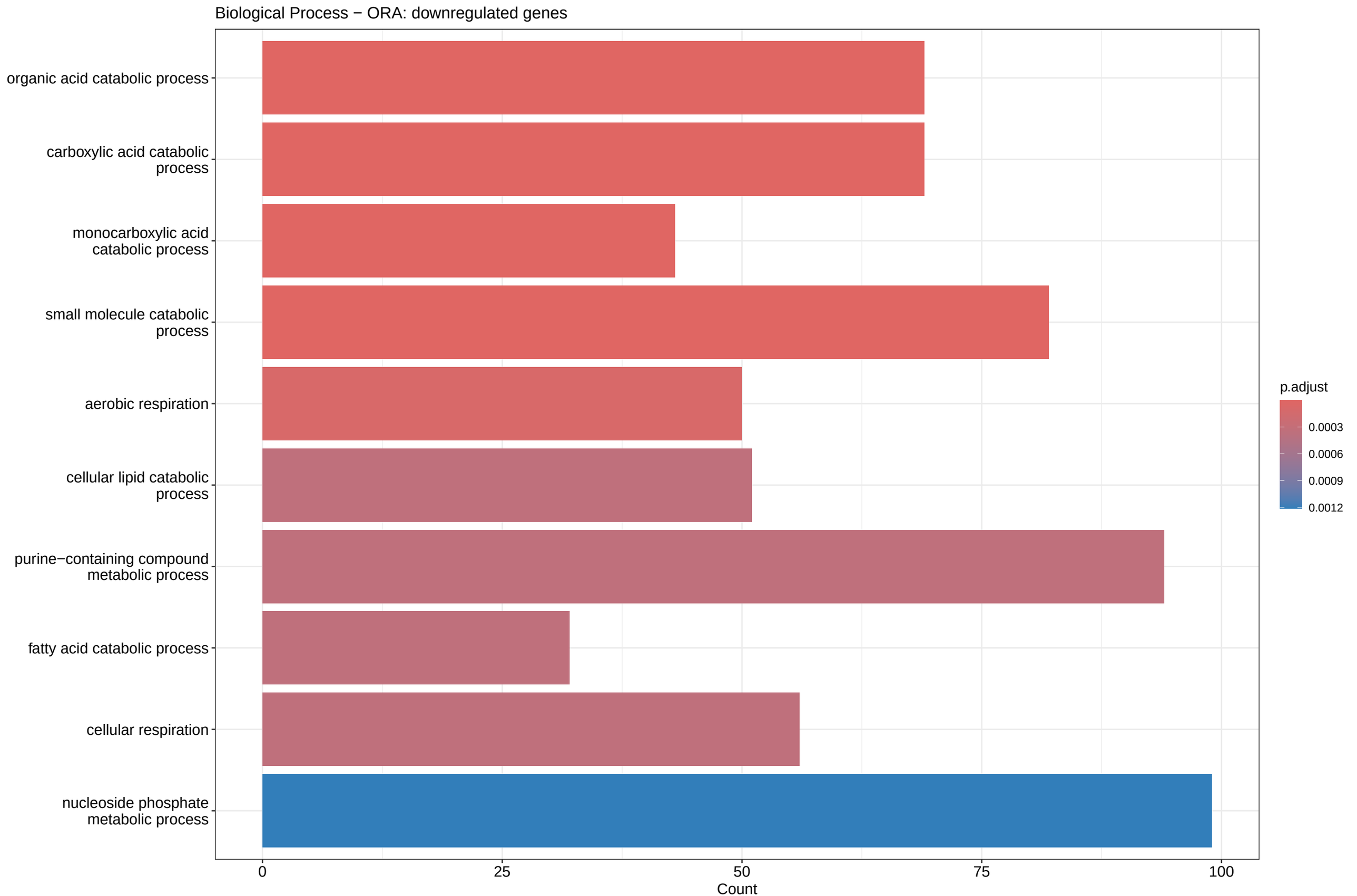


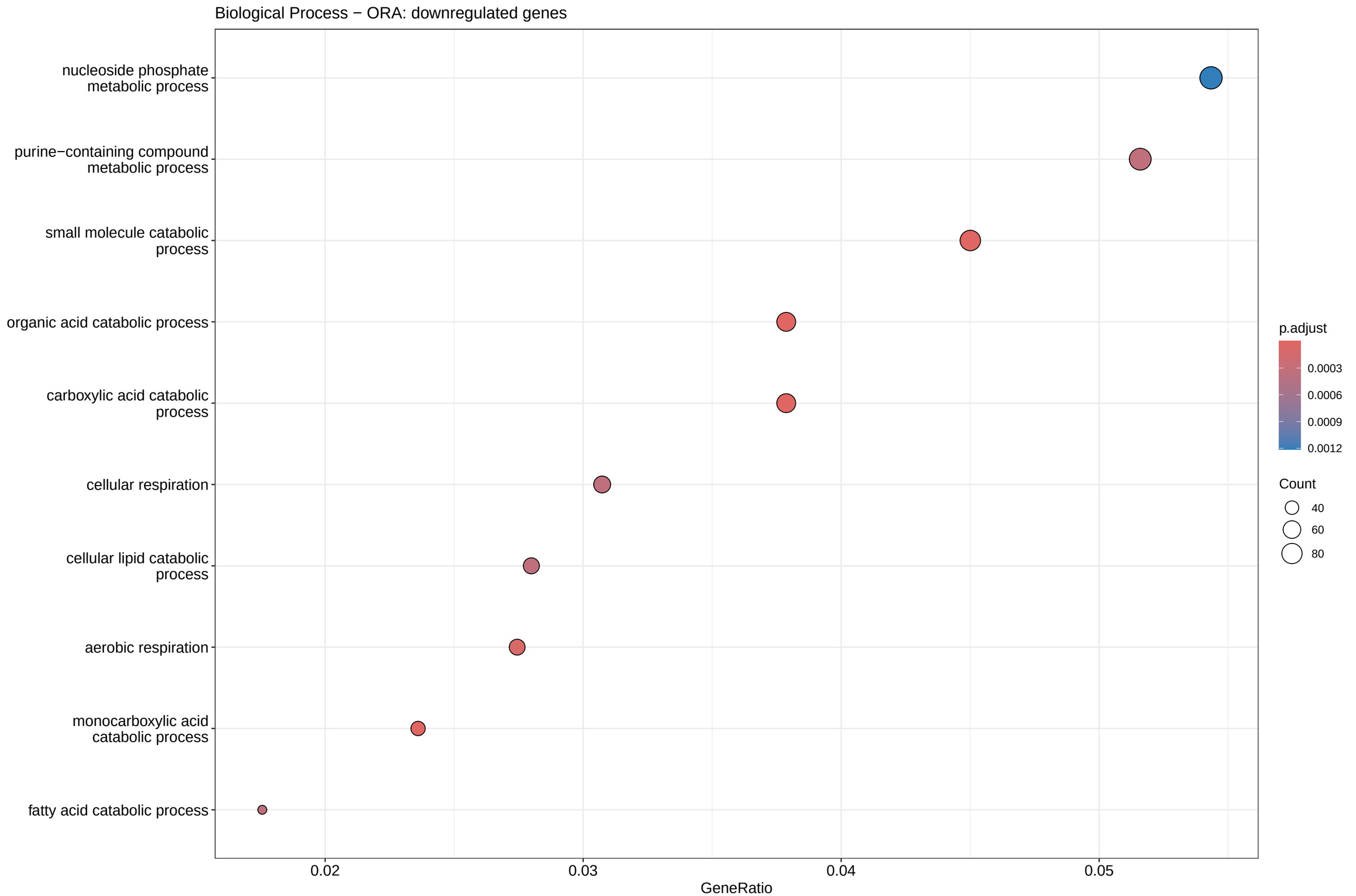

Biological Process − ORA: all genes
cytokine−mediated signaling pathway
regulation of cell−cell adhesion
CD4−positive, alpha−beta T cell differentiation
CD4−positive, alpha−beta T cell differentiation involved in immune response
positive regulation of cell−cell adhesion
positive regulation of cell activation
T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
alpha−beta T cell differentiation involved in immune response
T−helper cell differentiation
0
50
100
150
200
Count
p.adjust
1e−08
2e−08
3e−08

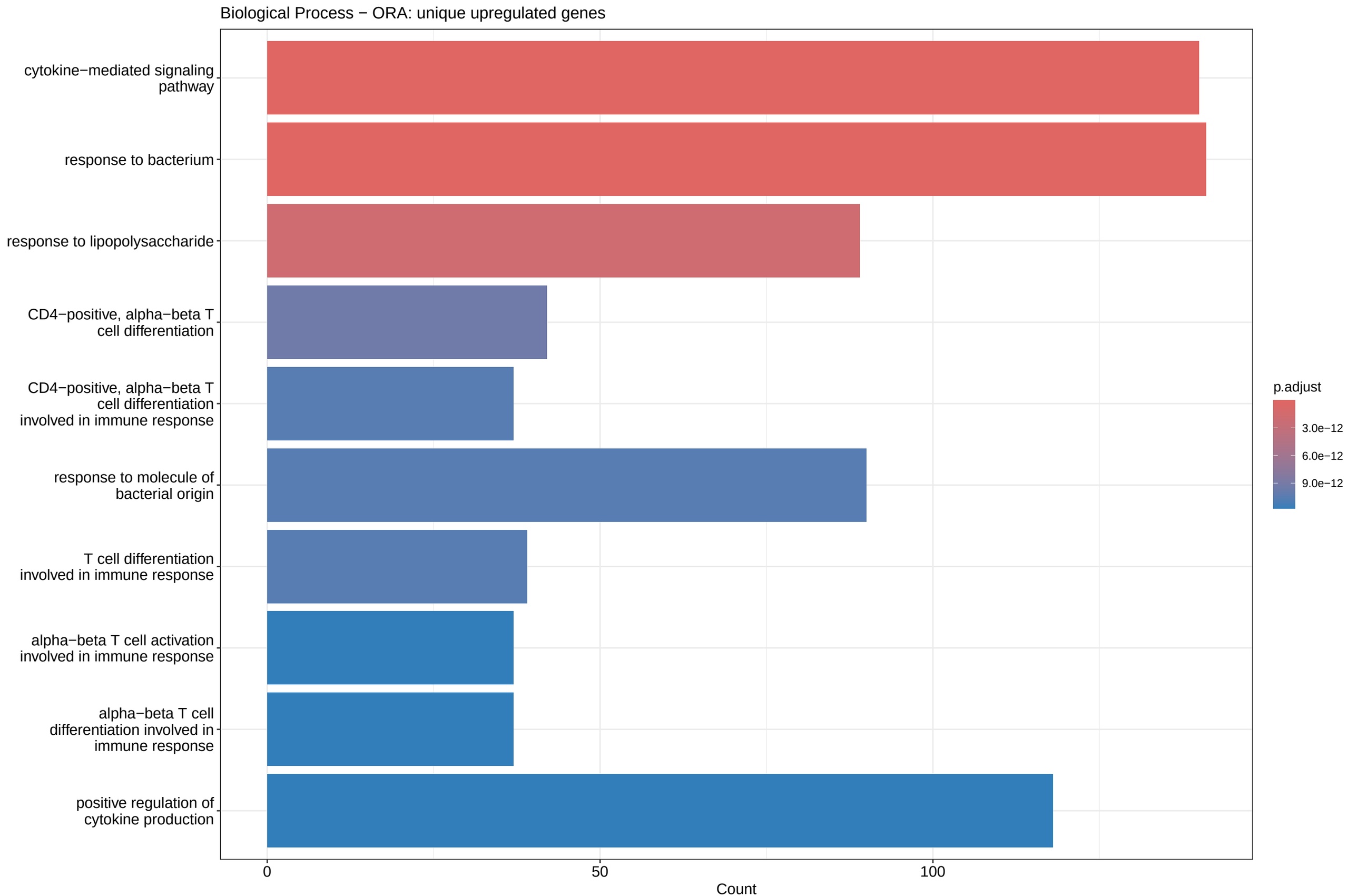

Biological Process − ORA: all genes
cytokine−mediated signaling pathway
regulation of cell−cell adhesion
positive regulation of cell activation
positive regulation of cell−cell adhesion
CD4−positive, alpha−beta T cell differentiation
T cell differentiation involved in immune response
CD4−positive, alpha−beta T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
alpha−beta T cell differentiation involved in immune response
T−helper cell differentiation
0.02
0.03
0.04
0.05
GeneRatio
p.adjust
1e−08
2e−08
3e−08
Count
50
100
150

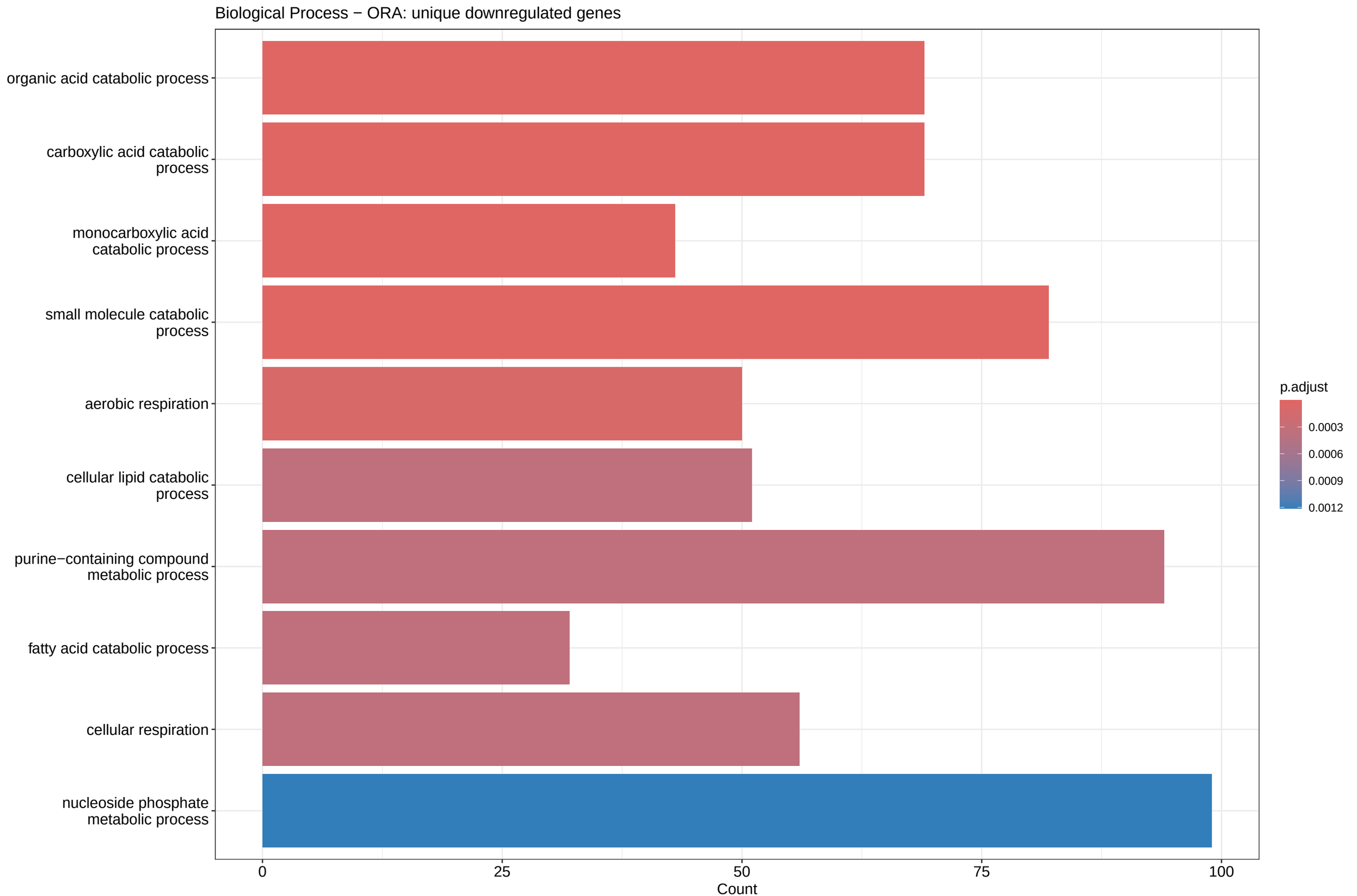

Biological Process − ORA: upregulated genes
cytokine−mediated signaling pathway
response to bacterium
response to lipopolysaccharide
CD4−positive, alpha−beta T cell differentiation
CD4−positive, alpha−beta T cell differentiation involved in immune response
response to molecule of bacterial origin
T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
alpha−beta T cell differentiation involved in immune response
positive regulation of cytokine production
0
50
100
Count
p.adjust
3.0e−12
6.0e−12
9.0e−12

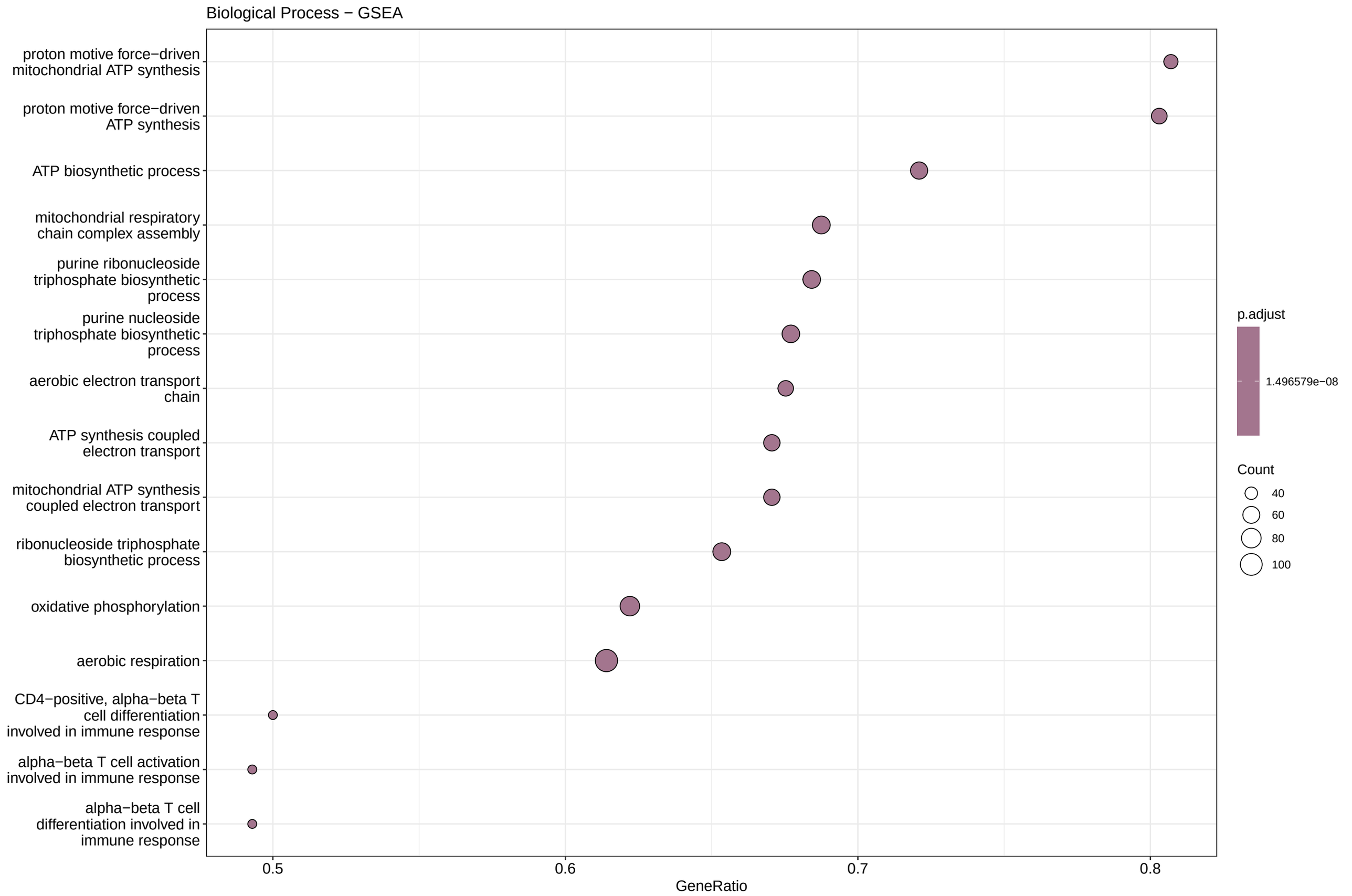

Biological Process − ORA: upregulated genes
response to bacterium
cytokine−mediated signaling pathway
positive regulation of cytokine production
response to molecule of bacterial origin
response to lipopolysaccharide
CD4−positive, alpha−beta T cell differentiation
T cell differentiation involved in immune response
CD4−positive, alpha−beta T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
alpha−beta T cell differentiation involved in immune response
0.02
0.04
0.06
GeneRatio
p.adjust
3.0e−12
6.0e−12
9.0e−12
Count
40
60
80
100
120
140

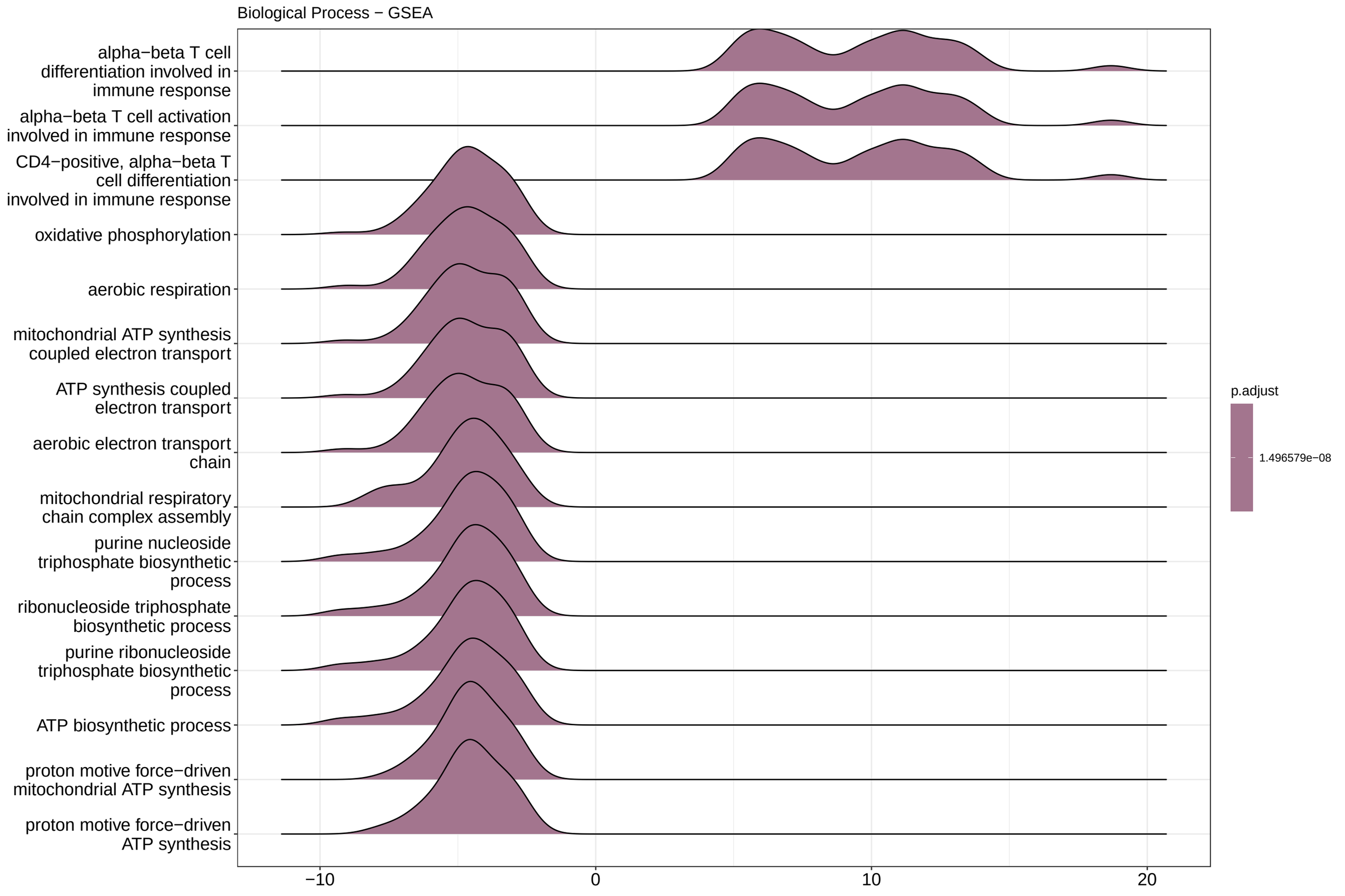

Biological Process − ORA: downregulated genes
organic acid catabolic process
carboxylic acid catabolic process
monocarboxylic acid catabolic process
small molecule catabolic process
aerobic respiration
cellular lipid catabolic process
purine−containing compound metabolic process
fatty acid catabolic process
cellular respiration
nucleoside phosphate metabolic process
0
25
50
75
100
Count
p.adjust
0.0003
0.0006
0.0009
0.0012


Biological Process − ORA: downregulated genes
nucleoside phosphate metabolic process
purine−containing compound metabolic process
small molecule catabolic process
organic acid catabolic process
carboxylic acid catabolic process
cellular respiration
cellular lipid catabolic process
aerobic respiration
monocarboxylic acid catabolic process
fatty acid catabolic process
0.02
0.03
0.04
0.05
GeneRatio
p.adjust
0.0003
0.0006
0.0009
0.0012
Count
40
60
80


Biological Process − ORA: unique upregulated genes
cytokine−mediated signaling pathway
response to bacterium
response to lipopolysaccharide
CD4−positive, alpha−beta T cell differentiation
CD4−positive, alpha−beta T cell differentiation involved in immune response
response to molecule of bacterial origin
T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
alpha−beta T cell differentiation involved in immune response
positive regulation of cytokine production
0
50
100
Count
p.adjust
3.0e−12
6.0e−12
9.0e−12


Biological Process − ORA: unique downregulated genes
organic acid catabolic process
carboxylic acid catabolic process
monocarboxylic acid catabolic process
small molecule catabolic process
aerobic respiration
cellular lipid catabolic process
purine−containing compound metabolic process
fatty acid catabolic process
cellular respiration
nucleoside phosphate metabolic process
0
25
50
75
100
Count
p.adjust
0.0003
0.0006
0.0009
0.0012


Biological Process − GSEA
proton motive force−driven mitochondrial ATP synthesis
proton motive force−driven ATP synthesis
ATP biosynthetic process
mitochondrial respiratory chain complex assembly
purine ribonucleoside triphosphate biosynthetic process
purine nucleoside triphosphate biosynthetic process
aerobic electron transport chain
ATP synthesis coupled electron transport
mitochondrial ATP synthesis coupled electron transport
ribonucleoside triphosphate biosynthetic process
oxidative phosphorylation
aerobic respiration
CD4−positive, alpha−beta T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
alpha−beta T cell differentiation involved in immune response
0.5
0.6
0.7
0.8
GeneRatio
p.adjust
1.496579e−08
Count
40
60
80
100


Biological Process − GSEA
alpha−beta T cell differentiation involved in immune response
alpha−beta T cell activation involved in immune response
CD4−positive, alpha−beta T cell differentiation involved in immune response
oxidative phosphorylation
aerobic respiration
mitochondrial ATP synthesis coupled electron transport
ATP synthesis coupled electron transport
aerobic electron transport chain
mitochondrial respiratory chain complex assembly
purine nucleoside triphosphate biosynthetic process
ribonucleoside triphosphate biosynthetic process
purine ribonucleoside triphosphate biosynthetic process
ATP biosynthetic process
proton motive force−driven mitochondrial ATP synthesis
proton motive force−driven ATP synthesis
−10
0
10
20
p.adjust
1.496579e−08